

Popular Generative Models for Synthetic Sound Generation

Introduction

Generative models are crucial in synthetic audio generation tasks, including bird sounds, speech synthesis, and environmental sound generation. While these models are not necessarily the state of the art (SoTA), they are widely used due to their effectiveness, popularity, and relative ease of use. Below is an overview of popular models and their performance.

Popular Generative Models for Audio Generation

The most commonly used models for generating synthetic sounds include:

- **WaveNet** [1]
- **Generative Adversarial Networks (GANs)** such as WaveGAN [2] and SpecGAN [3]
- **Diffusion Models** [4, 11]
- **Variational Autoencoders (VAEs)** [5]
- **Recurrent Neural Networks (RNNs)** such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs) [6, 10]
- **HiFi-GAN** [7]

Overview and Performance of Generative Models

WaveNet

WaveNet generates raw audio waveforms and is known for producing high-quality, natural-sounding audio. Despite its ability to model raw waveforms with great detail, it is computationally expensive and slow, particularly for long sequences [1].

GANs

Generative Adversarial Networks (GANs), especially Conditional GANs (cGANs), have been adapted for audio generation tasks. WaveGAN and SpecGAN are among the prominent examples. These models are generally faster than WaveNet for inference, but training can be unstable, requiring careful tuning of hyperparameters [2, 3].

SpecGAN

SpecGAN is a variation of GANs that generates audio spectrograms instead of directly working with raw waveforms. The spectrogram is then converted back into an audio waveform using inversion techniques like the Griffin-Lim algorithm. SpecGAN simplifies the generation process by focusing on the frequency domain, allowing it to more easily capture relevant audio features. It has been used in music and environmental sound generation, where generating spectrograms is sometimes easier than working with waveforms [3].

Diffusion Models

These models simulate a probabilistic diffusion process to generate realistic audio samples. While diffusion models show impressive results in audio generation, their training process is computationally intensive [4, 11].

VAEs

Variational Autoencoders (VAEs) are easier to train compared to GANs and WaveNet, and they produce reasonable quality audio outputs. However, VAEs often fall short in fidelity compared to models like GANs and WaveNet [5].

RNN-based Models

Recurrent Neural Networks (RNNs), particularly LSTMs and GRUs, are effective in capturing temporal dependencies in data. However, they typically struggle with high-fidelity audio synthesis and may not produce results comparable to WaveNet or GANs [6, 10].

HiFi-GAN

HiFi-GAN is a relatively recent model known for its ability to generate high-quality audio with faster inference time compared to earlier models like WaveNet. HiFi-GAN leverages GANs for efficient audio generation while maintaining high fidelity, making it one of the more effective models in modern synthetic sound generation tasks [7].

Performance Considerations

- For high-quality, natural-sounding audio, models like **WaveNet** and **Diffusion Models** are ideal due to their ability to model raw waveforms and probabilistic sampling.
- **GANs**, including **HiFi-GAN** and **SpecGAN**, provide a good balance of quality and computational efficiency, making them suitable for quicker outputs without severely compromising audio fidelity.
- **VAEs** offer an easier training process but typically produce lower fidelity compared to GANs and WaveNet.
- **RNNs** are suitable for tasks that require capturing temporal dependencies, but they are less effective for high-fidelity audio synthesis.

Conclusion

While newer models like HiFi-GAN offer faster and higher-quality audio generation, older models such as WaveNet and VAEs remain popular due to their ease of use and the quality they provide. Each model has its trade-offs between computational cost, training complexity, and output quality, and the choice of model depends on the specific application.

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